Ashlin Varughese Thomas

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Education

National Institute of Science Education and Research (NISER) Bhubaneswar, India
Integrated MSc. (Major: Physics, Minor: Mathematics) | ***CGPA: 9.79/10.0*** *2022 – Present*

- Currently enrolled in the penultimate semester; achieved the highest CGPA in the batch for academic years 2022-23, 2023-24, 2024-25, and 2025-26.

Citadel Residential School Kerala, India
AISSCE (Grade 12) | ***99.2%*** *(District Topper, Centum in Physics & Chemistry)* *2021 – 2022*

Research Interests

Core Areas: Condensed Matter Physics, Open Quantum Systems, Quantum Many-Body Physics, Non-Hermitian Physics and Topology, Mathematical Physics, Quantum Information, Nonlinear Dynamics.

Research Experience

Centre for Quantum Dynamics, Griffith University Australia
Research Intern | Advisor: Prof. Dr. Howard Wiseman & Dr. Lucas Ostrowski *Summer 2026*

- Developing a full quantum model of a cascaded amplifier system for circuit QED setups using superconducting circuits to surpass the standard quantum limit of laser coherence.
- Analyzing the coherence properties of output beams in the presence of losses, Kerr nonlinearity, and experimental imperfections.

Quantum and Complex Systems Group, Stockholm University Sweden
Research Intern | Advisor: Prof. Dr. Emil J. Bergholtz & Dr. Paolo Molignini *Winter 2026*

- Investigated the phase diagram of the extended Bose-Hubbard model in the presence of non-reciprocal hopping.
- Executed numerical simulations utilizing exact diagonalization via the ‘edith‘ package to analyze the interplay between the non-Hermitian skin effect and novel phase transitions.

Max Planck Institute for the Physics of Complex Systems Dresden, Germany
Research Intern | Advisor: Prof. Dr. Jan Michael Rost & Prof. Dr. Ulf Saalmann *Summer 2025*

- Engineered a unitary transformation framework to convert quantum Hamiltonians into ‘smooth images‘, creating optimized matrix representations for machine learning applications.
- Solved underlying non-convex optimization problems on the unitary manifold using Lagrange multipliers and analyzed structural robustness against noise.

Raman Research Institute (RRI) Bangalore, India
Research Intern | Advisor: Dr. Shovan Dutta (Collab: Prof. M. Haque & Dr. F. Fritzsch) *Summer 2024*

- Formulated a semiclassical quantization scheme for Lindblad dynamics in open quantum systems.
- Proposed a quasiprobability measure constructed from left and right eigenmodes of the non-Hermitian Liouvillian superoperator to interpret them as a set of coherences.

IISER Pune Pune, India
Research Intern | Advisor: Dr. Anisa Chorwadwala *Winter 2023*

- **Shape Optimization:** Studied isoperimetric problems and geometric functionals, focusing on configurations that minimize or maximize specific parameters like perimeter or volume under constraints.

National Institute of Science Education and Research (NISER) Bhubaneswar, India
Research Intern | Advisors: Dr. Sayantani Bhattacharyya & Dr. Anupam Pal Choudhury *Summer 2023*

- **Nonlinear Dynamics:** Modeled population interactions as a set of coupled differential equations and analyzed them using phase-plane analysis and numerical simulations based on nonlinear dynamical systems.
- **Planar Systems of Differential Equations:** Investigated the local and global behavior of autonomous planar systems of ordinary differential equations, exploring fixed points, linearization, and stability criteria.

Selected Projects & Talks

Turning Quantum Hamiltonians into Images | *Talk (Report, Slides)* Vikiran Physics Club

- Presented a mathematical framework for transforming finite-dimensional Hamiltonians into noise-resilient "smooth images" via unitary transformations, based on research conducted at MPIPKS Dresden.
- Explored applications of the methodology in visualizing quantum systems and contributing to quantum error mitigation and control protocols.

Harmonizing Classical and Quantum Mechanics | *Talk (Slides)* Vikiran Physics Club

- Presented a robust semi-classical approach to quantum mechanics utilizing the Wigner-Weyl transform to bridge the formalisms of classical phase space and quantum Hilbert spaces.

Magnon Bound States in 1D Heisenberg Chain | *Project (Report)*

- Investigated two-particle bound states using the Jordan-Wigner transformation and two-particle Green's functions, identifying exact criteria for localized "doublon" emergence.

Diagonalization of U(1) Covariant Lindbladians | *Project (Report)*

- Developed an exact combinatorial method using weighted directed graphs to find the eigensystem of Lindbladians covariant under U(1) symmetry.

Nonlinear Dynamics | *Project (Report)*

- Analyzed fixed points, stability, and bifurcations in linear and nonlinear systems utilizing numerical solving methods and phase-plane geometry.
- Constructed and evaluated a second-order nonlinear mathematical model to study the dynamics of interacting populations.

Google PageRank Algorithm | *Project (Report)*

- Explored the mathematical foundations of network search, modeling the internet using stochastic matrices and resolving computational challenges in algorithmic convergence.

Honors & Awards

INSPIRE Scholarship | *Department of Science and Technology, Govt. of India* 2022 – Present

- Awarded to the top 1% of students nationwide for pursuing advanced studies in natural sciences.

Academic Excellence Award | *NISER Bhubaneswar* 2023 – 2026

- Achieved the highest CGPA among all Int. MSc students for four consecutive academic years.

Teaching & Leadership Experience

Physics and Mathematics Tutor Kerala, India
Crispr Learning Pvt. Ltd. *2025 – 2026*

- Tutored advanced high school students for premier Indian research institute entrance exams, delivering weekly lectures on mechanics, electromagnetism, and calculus.

Project Group Supervisor NISER Bhubaneswar
Vikiran (Physics Club) *2023 – 2024*

- Led a team of students in applying concepts of dynamical systems; instructed weekly sessions on phase space, stability analysis, and numerical methods.

Technical Skills

Languages & Tools: Python, Mathematica, LaTeX, Linux/Bash
Spoken Languages: English (Fluent), Malayalam (Native), Hindi, Tamil